Chapter 11 Introduction to Communications

This chapter provides an introduction to communications and serial interfacing and practical information on selecting the protocol handshaking, cabling and other aspects of setting up the serial ports.

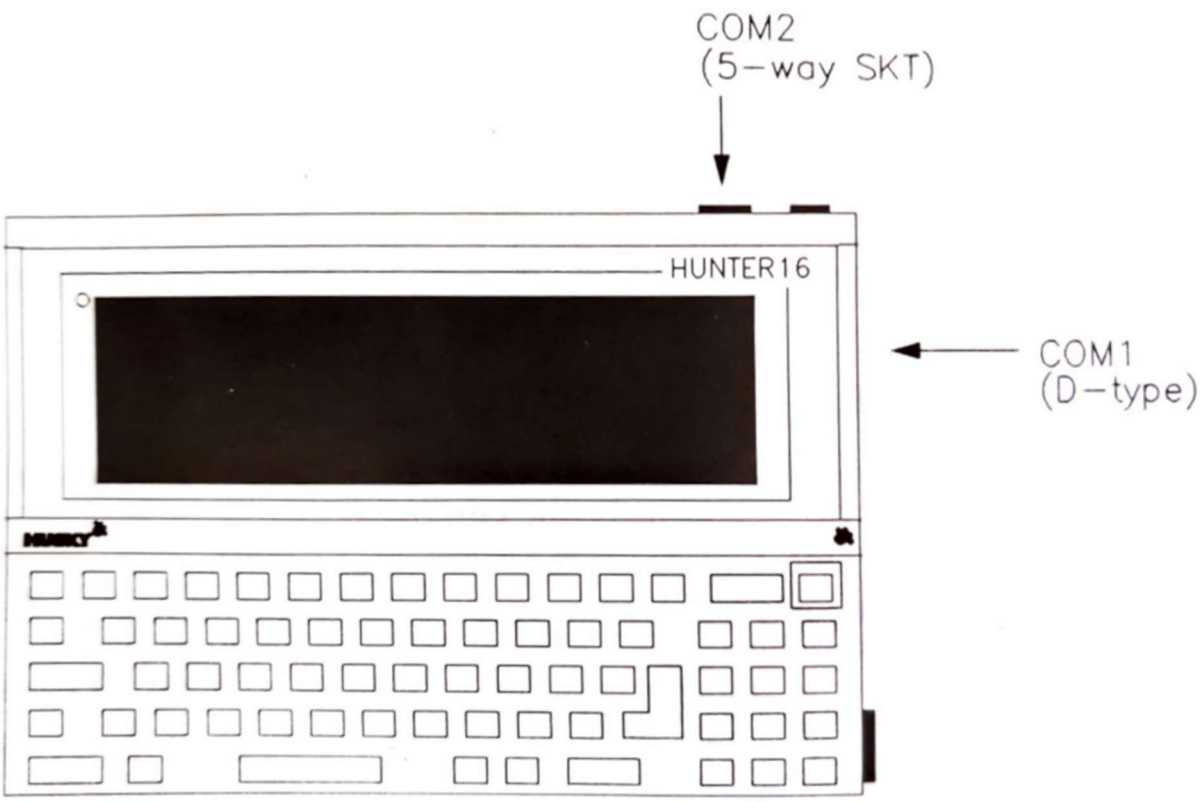
For information on using communications, see Chapter 12. For details of programming with communications, see Part 3. For more details of the communications options available, see Chapter 13 and your supplier.

11.1 Introduction

Communications facilities are especially important for portable computers, because they often rely on communications to exchange information or programs with remote computer systems.

Hunter 16 is unusual in providing extremely flexible and powerful communications as standard. This allows you to communicate with most other types of computer, peripherals, modems and other devices.

Hunter 16 has two industry-standard RS232/V24 serial communications ports:



These ports conform with the RS232/V24 specification, which is used for serial interfaces in virtually every computer system, terminal printer and modem. This allows Hunter 16 to 'plug 1n' to most other systems without modification, including microcomputer systems such as the IBM-PC.

Hunter 16 will normally be used as a self-contained computer, with the ports being used to connect suitable/peripherals such as printers or modems. The ports may also be used to a mainframe computer system, allowing you to use **Hunter 16** as a portable terminal.

Hunter 16s can also communicate with each other and other Husky computers, either side-by-side on the bench or over thousands of miles by telecommunication channels.

For maximum flexibility and data security a range of optional industry standard protocols and internal upgrades are available providing even more flexibility and sophisticated facilities such as error-correction.

NOTE: In this chapter, 'host' is used to refer to the system that is communicating with Hunter 16. It will typically be another computer printer, modem or terminal.

11.2 Communications packages

Hunter 16 has been designed to closely emulate IBM PC hardware, to give you access to the vast range of software available for that computer. This emulation includes the serial ports and communications.

Unfortunately, the IBM PC's communications facilities are rather primitive. To give you the best possible facilities, Husky Computers Ltd have developed a far superior communications package, which you can select when appropriate. This has the fallowing major advantages:

- Support for protocols (XON/XOFF and any optional protocols installed), providing more secure data transfer.
- Configurable handshaking, to suit a wide variety of devices and connection methods.
- Faster data transfer (up to 38,400 Baud) for efficient communications.
- Data buffering for reliable fast transfer.
- Compatibility with other Husky handheld computers (such as **Husky Hunter** and **Husky Hawk**).

This chapter is mainly concerned with setting up and using these extra facilities. For more details of standard IBM PC communications, refer to the *MS-DOS Programmer's Manual* and *MS-DOS User's Reference*.

Hunter 16 is provided with a simple file transfer utility, COM.COM. For details, see Chapter 12. You can even use a third party communications package, providing it will operate on IBM PC-compatible hardware. In this case, Husky communications will be completely ignored; the package will use the serial ports as it would on an IBM PC.

11.3 Background information

You do not need a detailed understanding of data communications to be able to utilise **Hunter 16's** powerful facilities. You simply need to know how to:

- Connect **Hunter 16** to the remote system.
- Set up **Hunter 16's** communications to suit the remote system.
- Use the communications facilities provided.

The following background information may however be of interest.

11.3.1 Buffering

Husky communications provides data buffering via invisible 'buffers' that effectively disconnect the application program from the outside world. The communications package is driven by interrupts, for maximum efficiency. This ensures a smooth flow of data and correct handling of communications even when **Hunter 16** is busy doing something quite different!

Because the communication package is interrupt driven, the execution of user programs is not affected by communication. However, the user program may slow down if the serial line is busy. **Hunter 16's** communication buffers are available whenever the machine is powered up, even if the application program is not requesting communication. Incoming messages will be received and held in the buffer, while any outgoing data remaining will continue to be sent down the line. Any protocols selected will continue to be observed.

11.3.2 Data Transmission

While using Husky communications, transmitted data is initially written to a 256 character transmit buffer (one for COM1 and one for COM2). Characters are removed from this buffer by the interrupt-driven communications package as and when the remote system is ready to receive them.

If the buff er becomes full, then the application is held up until the characters are sent (or a timeout occurs).

If a timeout or other communications failure is detected, it is reported and execution of the user program is suspended until the user acknowledges the error.

11.3.3 Data Reception

While using Husky communications, received data is placed initially in a 256 character wrap-round receive buffer (one for COM1 and one for COM2).

If the buffer becomes full, the protocol or handshaking which have been selected should prevent reception of further data. However if the remote system does not stop transmitting when requested to (ie does not respond correctly to the protocol or handshaking selected) the later characters overwrite the earlier characters.

An application reads data from this receive buff er. If there is sufficient data in the buff er, then it will be returned immediately. If there is insufficient data, **Hunter 16** will wait for it to arrive (or until a timeout occurs).

Reception of data is completely transparent to the application being run by **Hunter 16**. Data may be sent to Hunter 16 at any time and the communications protocols will function as configured.

If a timeout or other communications failure is detected, execution of the user program will be suspended until the user acknowledges the error. (See section 12.10.)

11.4 Connecting up

11.4.1 Main port (COM1)

To connect to the main port (COM1), you must use_ a standard male 25-way 'D' type connector. You can connect or disconnect at any time: you do not need to power down first.

This port is configured as a DTE. This means that you can connect direct to a DCE (such as most modems) by using a standard straight-through cable ie a cable with pin 1 wired to pin 1, pin 2 wired to pin 2, etc.

If the other device is a DTE (a printer, terminal, **Hunter 16**, or another computer), you must use a 'crossed' cable ie one with (at a minimum) pin 7 wired to pin 7, pin 2 to pin 3 and vice versa, pin 4 to pin 5 and vice versa.

Further details are provided in sections 11.4.3 and 11.7 for users who wish to wire their own cables.

NOTE: Never, never try to force a mating connector home. Check that the connector is the right way up. If the connector won't fit easily, it probably won't fit at all. FIND OUT WHY

11.4.2 Auxiliary port (COM2)

The auxiliary port (COM2) connects via a special 5-way connector known as a FISCHER. You are unlikely to find a standard cable to suit, so you should either obtain one from your usual supplier, or ref er to section 11.4.3.2 for wiring details. (Some of its pins are also duplicated on the 25-way D type; see section 11.4.3.1.)

To connect to the socket, you must align the red mark on the plug sleeve with the corresponding red mark on the socket. Push the plug in until the outer sleeve is flush with the **Hunter 16** casing.

You can connect or disconnect at any time: you do not need to power down first.

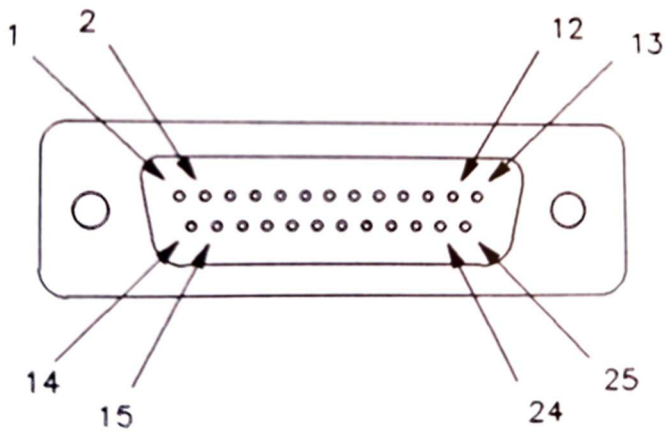
NOTE: Never, never try to force a mating connector home. Check that the connector is the right way up. If the connector won't fit easily, it probably won't fit at all. FIND OUT WHY

11.4.3 Pinouts

This section describes the pinouts for the two serial ports. Note that section 11.7 describes the cables you'll need to connect to most popular computers and peripherals.

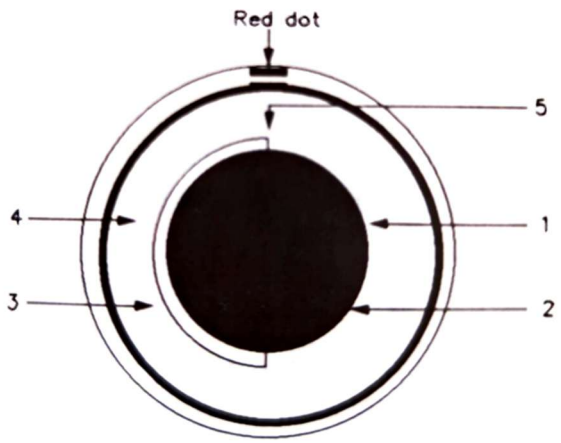
Remember that if you are using Husky communications, the behaviour of the handshaking lines is configured by the UTIL utility (or the application you are using).

11.4.3.1 COM1 (D type)



Pin	Description	Pin	Description
1	Protective Ground	14	TXDl (COM2)
2	Transmitted Data	15	Transmit Clock
3	Received Data	16	RXDl (COM2)
4	Request to Send	17	Receive Clock
5	Clear to Send	18	(Not used)
6	Data Set Ready	19	(Not used)
7	Signal Ground	20	Data Terminal Ready
8	Data Carrier Detect	21	(Not used)
9	(Not used)	22	Ring Indicator
10	+5V Power Out	23	(Not used)
11	Reserved	24	(Not used)
12	(Not used)	25	Charger In
13	CTS1 (COM2)		

11.4.3.2 COM2 (5- way FISCHER)



Pin	Description
1	Charger in
2	RX Data
3	Clear To Send
4	TX Data
5	Ground

The CTS input is also used for barcode wand input. This means that you cannot use COM2 handshaking and barcode input simultaneously, but this is unlikely to cause problems in practice.

11.5 Configuring Husky communications

When you reboot (or turn on for the first time), the default IBM communications is installed. To use Husky communications, you or the application must select it first.

If you are using an application written especially for **Hunter 16**, it will usually select and configure communications for you, automatically. In this case, you won't be required to intervene.

However, an application might present a menu for you to select from. In this case once you've selected the mode you require, it will configure communications without further action.

If you want to use Husky communications from MS-DOS or with an application that doesn't provide its own communications handling, you must configure communications first, by using the UTIL utility.

To do this, select MS-DOS command mode and type:

```
C:\UTIL
```

then select the communications menu option.

This displays the configuration for COM1. Use the [←] and [→] keys to move the highlight to the options you want to alter, then use [↑] and [↓] to adjust the displayed value as required.

When you have finished, press [ENTER] and repeat the process for COM2 Press [ESC] to return to MS-DOS command mode and the Husky communications package will be installed and configured as you have requested.

If you have installed one of the more complex optional protocols, an extra menu screen wiIl be displayed for you to configure. Full details will be supplied with the protocol.

If you simply want to examine the configuration, press [ESC] without using the cursor keys. The individual options will now be described after a brief overview of handshaking and protocols. For details of the operation of UTIL, refer to Chapter 8.

11.5.1 Handshaking

While you do not need to know how handshaking works in order to use Husky communications, you may find the following of interest.

Handshaking is a form of control provided by adding extra wires between the two communicating systems. These allow the two systems to communicate information such as 'I am ready to transmit' or 'I am ready for another character'.

Each handshaking signal is carried by a different wire and has a different name (CTS, RTS, DTR, etc). These names are often used interchangeably for the signals and for the wire or pin carrying it.

The main communications port includes a number of handshaking lines, while the secondary port includes a single handshake line (CTS). These are usually used when operating with a modem. The lines conform to the standard RS232 specification for use with modems, and may be configured to suit particular requirements. In general, a straight 25- way cable will operate with most modems.

Handshaking may also be used when communicating with slow devices (such as printers) which are connected direct to **Hunter 16**, or to allow **Hunter 16** to suspend data transmissions to it.

As handshaking involves extra wires it cannot be used over normal telephone lines, to control the remote system. Instead, handshaking is used to control the modem which connects **Hunter 16** to the telephone line. (A control mechanism similar to handshaking can be achieved by using a protocol, which includes control messages with the data being transmitted. See section 11.5.2.)

You can change the way **Hunter 16** produces and reacts to handshaking signals from **Hunter 16** applications or the UTIL utility. This makes communications rather easier than with systems which have fixed handshaking (with these, you often need to produce custom cables to 'trick' handshaking).

The individual handshaking lines are described in turn later in this chapter, with the single exception of Ring Indicator (RI). RI is an input which will turn **Hunter 16** on. It is often used by an auto-answer modem to turn on **Hunter 16** when a call is received so that the call can be processed.

11.5.2 Protocols

While you do not need to understand how protocols work in order to use communications, you may find the following information of interest.

Communications protocols are standard ways of organising data (for transmission, controlling transmission, detecting and handling errors.

A protocol transmits control information along the same path as used for data. Since it does not use extra wires, a protocol can be used over standard telephone lines, unlike handshaking.

Just like other configuration details, the protocols used on the transmitter and receiver must be the same.

The following protocols are provided as standard, for both ports:

- NONE
- XON/XOFF

The NONE protocol is suitable for transferring binary (8-bit) data; XON/ XOFF is not. Other protocols are also available as purchasable options.

These protocols will now be described in turn.

11.5.2.1 NONE

11.5.2.1.1 Reception

As its name implies, this selection does nothing to control the incoming data; transmission will normally be controlled by handshaking. The serial interface behaves like a simple Teletype. Characters are placed in the receive buffer when they are received, then removed when the application requests serial input.

If there is no room in the receive buffer, received characters will overwrite previously received characters. (This would normally be prevented by handshaking with the remote system.)

11.5.2.1.2 Transmission

Each character sent by the application is placed in the transmit buffer (if there is room). Characters are then removed and transmitted when the remote system is ready for them. Transmission would normally be controlled by handshaking.

If there is no room in the transmit buff er, the application will be made to wait for room.

11.5.2.2 XON/XOFF

11.5.2.2.1 Reception

If an incoming character causes the receive buffer to have 240 or more unread characters **Hunter 16** transmits an XOFF character (DC3, 13H or Control-S), which should prevent the host sending further characters. When there are only 100 characters left unread in the buffer, **Hunter 16** sends an XON character (DC1, 11H or Control-Q), which should tell the host to carry on transmitting.

Once **Hunter 16** has sent an XOFF, it continues sending them every three seconds until it is ready for more data. It then sends XONs

every three seconds until it receives more data. (This helps make sure that flow is controlled correctly, even if the host misses the first XON or XOFF.)

11.5.2.2.2 Transmission

While transmitting, **Hunter 16** continuously examines its input data line for XON or XOFF characters sent by the host.

If an XOFF or stream of XOFFs are received, Hunter 16 stops transmitting as soon as the character being transmitted is finished. It will only restart transmission when it receives an XON from the host.

By the very nature of this protocol, indeterminate handshaking situations may occur if any conditions are altered during transmission.

NOTE: When the host sends an XOFF, **Hunter 16** may send up to two more characters, depending on data link speed. This is due to the time taken to receive the XOFF and the possibility of just missing the start of the next character.

The host does not need to send an XON at the start of transmission.

11.5.2.3 Optional Protocols

To provide the fast, high integrity communication required by most mainframe computers, **Hunter 16** can be supplied with a number of advanced protocols.

For example, Husky Computers Ltd's HWK3780 is based on the IBM3780 synchronous protocol, but is asynchronous and uses the ASCII character set rather than the EBCDIC character set. This protocol includes error detection and automatic retries and binary data capability. It has been developed for trouble-free communications with other Husky computers and Husky Communications Managers over normal telephone lines.

Other protocols may also be available if additional hardware has been fitted, such as a full emulation of bisynchronous IBM3780 protocol.

If you select one of these protocols by using UTIL, you must supply additional information to tell **Hunter 16** how to configure the protocol. You do this by selecting options from an extra menu screen which is displayed when you press [ENTER].

For full details of the available protocols contact your supplier.

11.5.3 Port set-up options

The port set- up options will now be described, in alphabetical order. Except where noted otherwise, all options apply to both COM1 and COM2.

BAUD RATE

Use

To set the transmit and receive baud rates (speeds) to suit the connected device.

Notes

The available baud rates for COM1 and COM2 are:

110	1200	9600
150	2400	19200
300	4800	38400
600		

Note that external devices may not be able to handle the highest Baud rates.

CTS

Use

To control **Hunter 16's** response to the CTS (Clear To Send) handshake input while transmitting. (Pin 5 on D type for COM1, pin 21 on D type or pin 3 on 5-way FISCHER for COM2)

Notes

Setting	Result
y	**Hunter 16** will only transmit data if CTS is active (positive). CTS is tested at the start of each character. If CTS goes inactive part way through transmitting a character, that character will be completed before transmission stops.
n	CTS is ignored.

DCD

Use

To control **Hunter 16's** reaction to the DCD (Data Carrier Detect) handshake input on COM1. (Pin 8 on D type)

Notes

Setting	Result
y	**Hunter 16** will receive data normally only while DCD is active (positive).
n	DCD is ignored.

DSR

Use

To control Hunter 16's reaction to the DSR (Data Set Ready) handshake input on COM1, while transmitting or receiving. (Pin 6 on D type)

Notes

The connected device (usually a modem) can use this signal to tell **Hunter 16** when it is ready to operate.

Setting	Result
y	**Hunter 16** will transmit or receive only while DSR is active (positive).
n	DSR is ignored.

DTR

Use

To control production of the DTR (Data Terminal Ready) handshake output on COM1, while transmitting or receiving. (Pin 20 of D type)

Notes

This. signal is provided to allow you to use auto-answer modems that require DTR to be present before generating carrier, or the **Hunter 16 Printer Sidebox**, which requires DTR to print.

Setting	Result
Yes	DTR goes active (positive) if Hunter 16 is ready to receive or transmit.
No	DTR remains inactive.
On	DTR permanently on.

ECHO

Use

To control whether characters will be displayed on the screen as they are transmitted.

Notes

Data transmission and reception are not affected by echoing and the displayed characters are not passed to user programs or entered in the receive buffer.

ECHO is usually used to mimic full duplex operation with a half duplex remote system. If the remote system is operating in full duplex mode, selecting ECHO will display each character twice!

Setting	Result
y	Characters are echoed as they are transmitted.
n	Characters are not echoed.

Echoing a non-text file is not recommended.

LF

Use

To control automatic transmission of a LF character (decimal 10, hexadecimal 0AH) after each CR (decimal 13, hexadecimal 0DH) when transmitting.

Notes

This option is mainly provided for use with printers or other remote systems that require an LF after a CR to start a new line.

Setting	Result
y	A LF will be inserted after each CR transmitted even- 1 if there is already a LF in the text.
n	Extra LFs will not be inserted, although any LFs in the data will be transmitted as normal.

NOTE: When sending binary data (such as a machine code program) set LF to 'n' to avoid corrupting the data.

Also note that with LF on, you are adding characters to the transmitted data, so the length will be greater and mechanisms such as checksums will be misled.

LENGTH

Use

To set the number of data bits transmitted or expected when receiving.

Notes

Setting	Result
7	Seven data bits
8	Eight data bits

NOTE: If transmitting binary files, set LENGTH to 8.

Also note that **Hunter 16** does not support 5-bit and 6-bit data lengths, which the IBM PC does. This is unlikely to cause problems, as these data lengths are very rarely used.

NULL

Use

To control whether **Hunter 16** inserts NULL characters (decimal0, hexadecimal 00H) after each CR, when transmitting.

Notes

This option is mainly provided for use with printers that take a significant time to perform a Carriage Return operation and don't have handshaking or buffering. NULLs are not printed, they simply give the printer time to perform the CR function.

Setting	Result
0	No NULLs are added.
1..20	1 .. 20 NULLs are added.

If LF is set to 'y', any NULLs will be sent after the CR LF sequence.

NOTE: If transmitting binary files, set NULL to 0 to avoid corrupting the data.

PARITY

Use

To select the data parity to add to transmitted data, and the data parity checking that will be used to check received data.

Notes

When transmitting, PARITY has the fallowing significance:

Setting	Result
None	A parity bit is not added.
Odd	A parity bit of 0 or 1 is added to produce an odd number of logical 1's in the data bits.
Even	A parity bit of 0 or 1 is added to produce an even number of logical 1's in the data bits.

When receiving, PTY has the following significance:

Setting	Result
None	A parity bit is not expected. If parity bits are sent, they will produce data framing errors.
Odd	The data is checked for an odd number of logical 1's (including the parity bit). If the check succeeds, the character is returned to the calling program. If the check fails, the character is not returned and a parity error is flagged.
Even	The data is checked for an even number of logical 1's (including the parity bit). If the check succeeds, the character is returned to the calling program. If the check fails, the character is not returned and a parity error is flagged.

Parity errors on receive may be ignored, display a communications error message (see section 12.10), or be handled by your application (see Part 3).

Note that **Hunter 16** does not support a fixed parity of 1, unlike the IBM PC. This is unlikely to cause problems as fixed parity is very rarely used.

PROTOCOL

Use

To select the protocol to use for communications.

Notes

The protocols available on **Hunter 16** are:

- NONE
- XON/XOFF

Other protocols may also be available if they have been installed. Selecting one of these more complex protocols will display an extra menu screen, which you must use to configure it. See section 11.5.2 for further details of these protocols.

RTS

Use

To control production of the RTS (Request To Send) handshake output on COM1 while transmitting or receiving data. (Pin 4 of COM1 D type).

Notes

Hunter 16 uses this signal either to tell a remote device that is ready to send data, or to prevent further reception when its receive buffer is full.

Setting	Result
OFF	The RTS line is permanently active (positive).
HOLD	The RTS line is normally active (positive) and only goes inactive (negative) when **Hunter 16's** receive buffer is full. This allows **Hunter 16** to hold up a transmitting device (by lowering the RTS line) until it has removed enough characters from the receive buffer to allow the transmitting device to send more data (by raising the RTS line again).
TRUE	This mode can be used when communicating with another Husky computer, via a crossed cable. Normally, RTS would be connected to the transmitting device's CTS input. This mode is mainly used with a half duplex modem or similar device, when sending data in blocks and relying on the protocol for flow control. When **Hunter 16** wishes to transmit data, it first of all raises RTS to active (positive), then waits for a reply on the CTS line (if enabled) before sending the data in the transmit buffer. RTS remains active until all the data has been sent and the buffer is empty, then Hunter 16 returns RTS to the inactive state (negative).

RX T/O

Use

To set the timeout period for data reception.

Notes

The timeout is initiated by a request for serial input by an application. If a character is not read from the receive buffer within the specified time limit (because it is empty) a communications error will result.

Reception of any character, or noise, will reset the timeout.

Note that if no attempt is made to read the input communications buffer, this timeout will not occur.

Setting	Result
0 1..60	No timeout Timeout after 1 to 60 seconds

NOTE: If an optional protocol is being used which has its own timeouts, disable the timeout.

SERIG

Use

To filter a particular character from received data.

Notes

This tells **Hunter 16** to remove a particular ASCII character from serial input ie to receive it normally but not place it in the receive buffer, so the character is never passed to the application.

This option is provided for use with computer systems that transmit abundant spurious characters (such as NULL, LF, XON or DEL) along with the real data. To save the application having to filter these unwanted characters, use SERIG.

Setting	Result
0-127 OFF	Filter the specified ASCII code from received data. Do not filter any characters.

To set SERIG to 'off', increment past 127 or decrement past 0.

NOTE: Do not use SERIG with binary data or protocols other than NONE or XON/XOFF.

TX T/O

Use

To set the timeout period for data transmission.

Notes

The timeout is initiated by a request for serial output by an application. If the application fails to put a character in the transmit buffer (because it is full) within the specified time limit, a communications error will occur.

If no attempt is made to transmit data, this timeout will not occur.

Setting	Result
0 1..60	No timeout Timeout after 1 to 60 seconds

NOTE: If an optional protocol is being used which has its own timeouts, disable the TX timeout.

11.6 Configuring IBM communications

Standard IBM communications will be selected by any standard application or command that initialises commun1cattons. The default configuration is as follows:

	COM1	COM2
Speed:	2400	2400
Data bits:	8	8
Parity:	N	N
Stop bits:	1	1
Retry on timeout:	N	N

(All handshake lines are inactive and interrupts are disabled.)

You can alter all these settings from MS-DOS command mode, by using the standard MS-DOS MODE utility. For example, to select 4800 Baud, 8 data bits, odd parity, I stop bit and no timeout for COM2, type:

```
MODE COM2 48,0,8,1,P
```

Note that the following IBM PC-compatible settings are not supported:

Data bits:	5, 6
Parity:	1 (fixed)
Stop bits:	1.5

These limitations are unlikely to affect you in practice, as these settings are very rarely used.

NOTE: For details of the MODE utility refer to the *MS-DOS User's Reference*.

11.7 Example communications set-up

This section describes the cables and configurations you will need to connect the main serial port to a number of popular devices, in order to transfer binary or ASCII (text) data.

11.7.1 Another Hunter 16

11.7.1.1 Cable

To connect one Hunter 16 direct to another you need a simple 'crossed cable':

First Hunter 16 (25-way female)		Second Hunter 16 (25-way female)
2	----------	3
3	----------	2
4	----------	5
5	----------	4
7	----------	7

11.7.1.2 Configuration

As the two computers are connected together by a short cable, you can use the fastest speed available, simple CTS/RTS handshaking and parity checking. There is no need for a protocol, as the handshaking controls data flow and parity checking provides reasonable security.

Configure both **Hunter 16s** as follows:

Option	Setting	Comments
Baud rate:	38400	As fast as you can, to save time.
Protocol:	None	None needed.
Length:	8	Binary data.
Parity:	Even	Allows data integrity checking.
Stop bits:	1	Minimum required to save time.
LF:	n	Adding LFs would corrupt binary data.
Null:	0	Adding NULLs would corrupt binary data.
RTS:	Hold	Handshake output when receiving and no more room. (Connected to transmitter's CTS.)
DTR:	n	Not needed.
CTS:	y	Handshake input when transmitting, allowing transmission to proceed. (Connected to receiver's RTS.)
DSR:	n	Not needed.
DCD:	n	Not needed.
Echo:	n	There's no point viewing binary data!
TXt/o:	30	As there's a direct connection, you only need a short timeout, but a longer time makes operation easier.
RXt/o:	10	A fairly short timeout, as this is used to indicate the end of the received file.
Serig:	off	Filtering bytes would corrupt binary data.

11.7.2 Husky Hunter

11.7.2.1 Cable

As section 11.7.1.

11.7.2.2 Configuration

Configure both computers as described in section 11.7.1, but as the maximum speed that **Hunter** can support is 4800 Baud, select this speed on both systems. On **Hunter**, data length and parity are combined into a single option: select a PTY setting of 8 + ev.

The **Hunter** COM1 port is configured by using the COMS command.

11.7.3 Husky Hawk and Hawk 8/16

11.7.3.1 Cable

The cable required is as described in section 11.7.1, but the **Husky Hawk** end should be a 25-way male.

11.7.3.2 Configuration

Configure the two computers as described in section 11.7.1.

The **Husky Hawk** COM1 port is configured by using the COM1 command.

The **Husky Hawk** 8/16 COM1 port is configured by using the SETCOM command.

11.7.4 External modem

11.7.4.1 Cable

Most modems are configured as DCEs, so you can use a 'straight' cable 1e wired pin 1to pin 1, pin 2 to pin 2, etc.

11.7.4.2 Configuration

You must configure the handshaking and data format to suit the modem, but select a protocol to suit the remote computer.

For example, to suit a typical modem, configure **Hunter 16** as follows:

Option	Setting	Comments
Baud rate:	*	To suit the modem.
Protocol:	*	To suit the remote system.
Length:	8	Binary data.
Parity:	*	To suit the modem.
Stop bits:	*	To suit the modem.
LF:	n	Adding LFs would corrupt binary data.
Null:	0	Adding NULLS would corrupt binary data.
RTS:	True	Handshake output when Hunter 16 wants to start transmission. (Modem replies on CTS.)
DTR:	y	May be needed to turn on modem.
CTS:	y	Handshake input from modem, when **Hunter 16** asks to start transmitting (by raising RTS).
DSR:	y	Input to indicate that modem is ready to operate.
DCD:	y	Input to indicate received data is valid.
Echo:	n	There's no point viewing binary data.
TXt/o:	30	To help detect any problem between **Hunter 16** and the modem.
RXt/o:	30	To help detect any problem **Hunter 16** and the modem.
Serig:	off	Filtering bytes would corrupt binary data.

Configure the modem interface speed, parity, data and stop bits as **Hunter 16.** Configure the modem line speed to suit the remote modem.

11.7.5 IBM personal computers

11.7.5.1 Cable (25- way)

IBM PCs, PS/2s and compatibles are normally fitted with a 25-way socket, usually female, but sometimes male. This requires a cable wired as follows:

Hunter 16 (25-way female)		IBM PC-AT (25-way female or male)
2	----------	3
3	----------	2
4	----------	6
5	----------	4
6	----------	5
6	----------	20
7	----------	7

Note that pins 5 and 20 in the IBM end of the cable are both connected to pin 6 in the Hunter 16 end. This will typically be achieved by linking them at the IBM end of the cable.

If a 9-way socket is fitted, you will require a cable wired as shown in section 11.7.5.2.

11.7.5.2 Cable (9-way)

IBM PC-ATs and compatibles may be fitted with 25-way sockets, or 9-way sockets. If a 25-way socket if fitted, you will require a cable wired as shown in section 11.7.5.1.

If a 9-way socket is fitted, the following cable is required.

Hunter 16 (25-way female)		IBM PC-AT (9-way female or male)
2	----------	2
3	----------	3
4	----------	6
5	----------	7
6	----------	8
6	----------	4
7	----------	5

Note that pins 4 and 8 in the IBM end of the cable are both connected to pin 6 in the Hunter 16 end. This will typically be achieved by linking them at the IBM end of the cable.

11.7.5.3 Configuration

Configure **Hunter 16** as follows:

Option	Setting	Comments
Baud rate:	9600	As fast as the IBM can go.
Protocol:	None	None needed.
Length:	8	Binary data.
Parity:	Even	Allows data integrity checking.
Stop bits:	1	Minimum required to save time.
LF:	n	Adding LFs would corrupt binary data.
Null:	0	Adding NULLS would corrupt binary data.
RTS:	Hold	Handshake output when receiving and no more room. (Connected to transmitter's CTS.)
DTR:	n	Not needed.
CTS:	y	Handshake input when transmitting, allowing transmission to proceed. (Connected to receiver's RTS.)
DSR:	n	Not needed.
DCD:	n	Not needed.
Echo:	n	There's not much point in viewing binary data.
TXt/o:	0	Not needed, but could set to 30.
RXt/o:	0	Not needed, but could set to 30.
Serig:	off	Filtering bytes would corrupt binary data.

Configure the IBM as follows:

Option	Setting
Speed:	9600
Data bits:	8
Parity:	E
Stop bits:	1
Retry on timeout:	N

11.7.6 Serial printer

11.7.6.1 Cable

To connect to a serial printer, you'll need a simple 'crossed cable':

```
Hunter 16                Printer
          2 ---------------- 2
          3 ---------------- 3
          4 ---------------- 5
          5 ---------------- 4
          7 ---------------- 7
```

Very few printers transmit data back to the computer, so you don't really need to connect the printer's TXD and CTS pins (pin 2 and pin 5).

If you are willing to use a slow Baud rate, you may even be able to dispense with all handshaking and use the following 2-wire cable:

```
Hunter 16                Printer
          2 ---------------- 3
          7 ---------------- 7
```

11.7.6.2 Configuration

As the printer is connected by a short cable, you can use the fastest speed available and simple CTS/RTS handshaking.

To show some of the versatility provided by configurable communications, I will assume that the printer does not have a buffer, perform a LF automatically when it receives a CR character, or handshake properly, but does support the XON/XOFF protocol.

Configure **Hunter 16** as follows:

Option	Setting	Comments
Baud rate:	*	Set to suit printer. (As fast as you can, to save time.)
Protocol:	XON/XOFF	Used to supplement handshaking.
Length:	7	ASCII data.
Parity:	*	Set to suit printer.
Stop bits:	*	Set to suit printer
LF:	y	Adding LFs after CRs to make the printer advance the paper correctly.
Null:	20	Needed to give printer time to perform CR LF.
RTS:	-	Irrelevant.
DTR:	n	Not needed.
CTS:	y	Handshake input when transmitting, allowing transmission to proceed. (Connected to printer's RTS.)
DSR:	n	Not needed.
DCD:	n	Not needed.
Echo:	y	To monitor progress.
TXt/o:	10	This timeout will allow you to monitor problems such as paper out or off-line.
RXt/o:	-	Irrelevant as printers don't usually transmit data!
Serig:	off	Irrelevant.

Configure the printer as follows:

Set the baud rate, parity, data bits and speed as set on **Hunter 16**, and select XON/XOFF protocol.

(Or, if you want to use the simple 2-wire cable, set CTS to 'n' and select a Baud rate which is significantly slower than the printer's printing speed. For example, if the printer prints at 40cps, select a speed of 300 Baud or less.)